ENERPRO

Twelve-Pulse Firing Packages

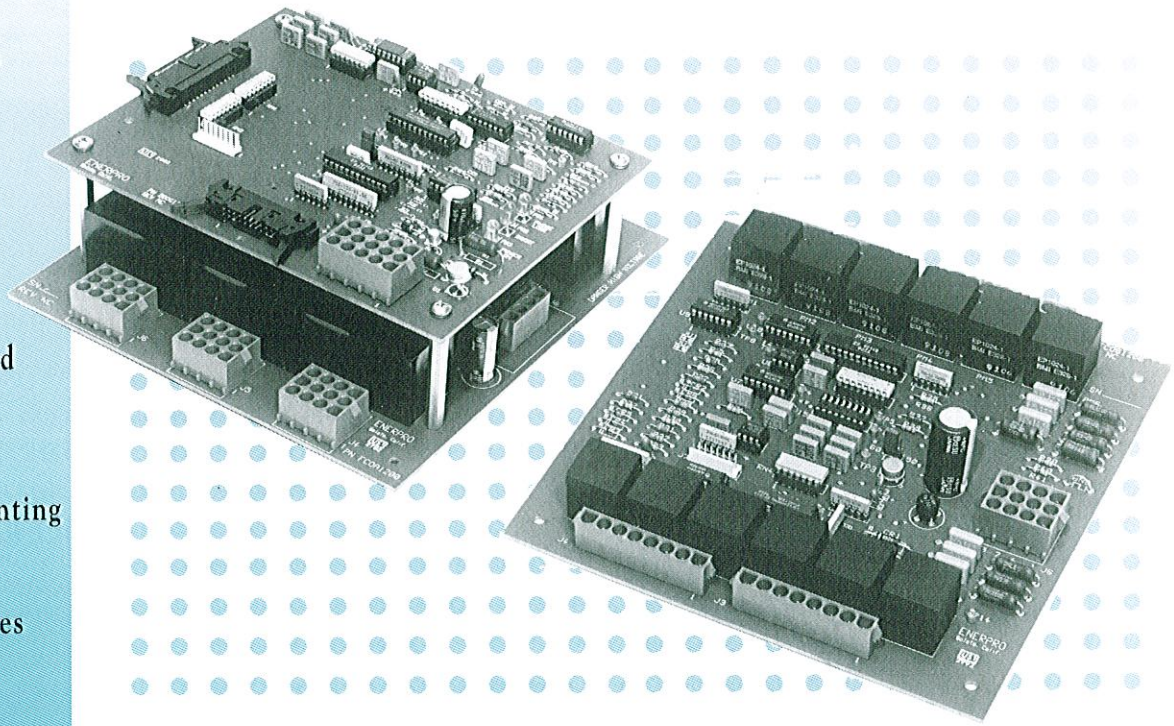
For 12-Pulse Converters or Controllers

Virtually Eliminates 5th and 7th Harmonics

Complete, Self-Contained Control Package

Designed for Panel Mounting

Complete SCR Assemblies also Available



The ENERPRO Model FCOG1200 Firing Board and the FCOP1200 Firing Package supply delayed, hard-firing gating pulses for parallel or series connected 12-pulse converters or ac controllers.

The twelve gate outputs are spaced at 30° intervals as required to eliminate the 5th and 7th harmonics of the ac input current. Adjustment is provided to compensate for transformer variations and firing circuit timing to achieve minimum phase imbalance and dc ripple voltage.

Standard features include power-on reset, soft-start and soft-stop, phase loss inhibit and instant inhibit. Gate delay angle transfer function is configurable for your application.

In addition, Enerpro engineers can design and produce a complete system to your specifications, including SCRs, heat sinks, regulator boards and firing package.

600 Vac and 900 Vac Models

The Model FCOG1200 Firing Board is designed to handle mains voltage up to 600Vac. It contains the thyristor gate delay determinator logic (including Enerpro's proprietary LSI circuitry) and the thyristor gate pulse transformers and associated circuitry on a single board.

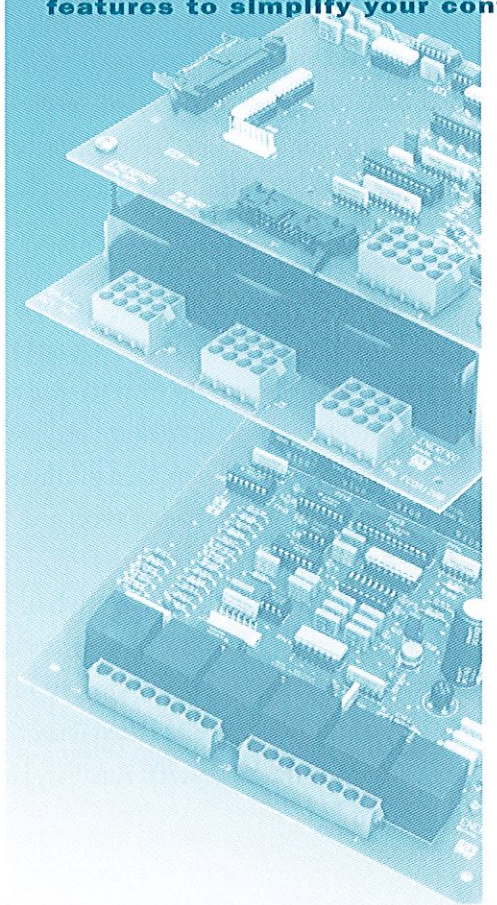
The Model FCOP1200 is a two-board firing package capable of handling input voltages up to 900Vac. It is made up of the DDOG1200 Gate Delay Determinator Board and the FCOA1200 Auxiliary Firing Board.

POWER ELECTRONICS ENERPRO CONTROL SPECIALISTS

ENERPRO

12-PULSE FIRING PACKAGES

The Enerpro FCOG1200 (single board/600Vac) and FCOP1200 (dual board/900Vac) twelve-pulse firing packages employ a six-phase, phase-locked-loop to achieve precise 30°, equidistant gating and excellent bandwidth. These unique, power quality enhancing power control devices employ Enerpro's proprietary firing circuitry incorporating LSI technology, first introduced in 1983 and firmly established as the North American industry standard. And they are designed with built-in features to simplify your controller or converter design.



POWER-ON RESET

The power-on reset feature prevents unscheduled thyristor gating when 3ø power is applied to the thyristors or when 1ø power is applied to the circuit board power supply. The power-on reset circuit activates the soft-start circuit.

SOFT-START & SOFT-STOP

The soft-start circuit overrides the gate delay angle command. It is activated by contact closure input or via the power-on reset feature. Gating commences at the maximum delay angle limit and ramps down to the commanded delay angle at a rate determined by the soft-start time constant. The soft-stop feature, when activated by contact closure, causes the gate delay angle to ramp up from the command angle to the maximum delay angle limit before gating is inhibited.

PHASE LOSS INHIBIT

The phase loss circuit removes thyristor gate drive if loss of phase voltage due to a blown fuse or other cause is detected. When the mains voltage is restored, the soft-start circuit is engaged.

INSTANT INHIBIT

Thyristor gating is instantly enabled or inhibited upon contact closure input.

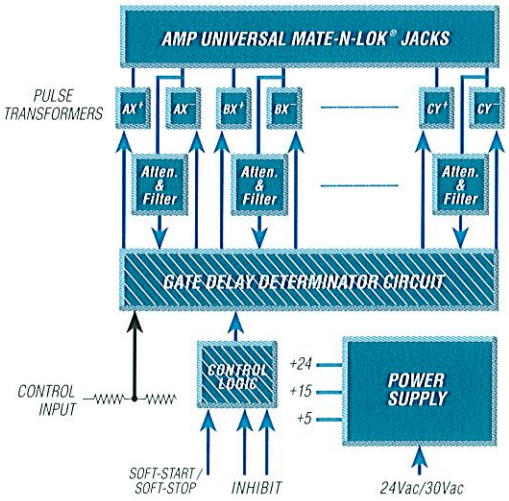
OPERATIONAL DETAILS

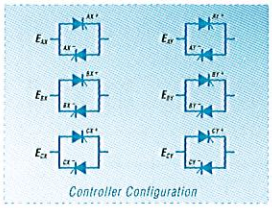
GATE DELAY TRANSFER FUNCTION

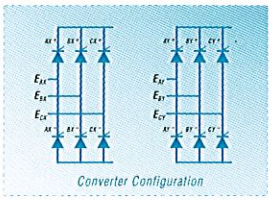
The thyristor gate delay is controlled by a 0 to 5 Vdc input signal. Gate delay angle varies in negative proportion to the command voltage. Minimum and maximum delay angles are selected for the specified application. An input resistor network is provided to accommodate offset voltage or current input control signals.

PHASE REFERENCE SENSING

Low level phase reference signals are produced by precision resistor attenuator networks and low-pass filters on the two sets of 30° phase-displaced 3ø mains voltages. The mains voltages are accessed at the thyristor gate pulse transformer windings connected to each of the six phases. The attenuated mains voltage reference signals are processed by low pass filters with 6dB attenuation and 60° phase shift at the mains frequency. These filters remove distortion from the reference signals caused by non-linear loading of the mains by the thyristor converter/controller. The reference signals are then selected by means of CMOS switches to cancel the 60° filter phase shift and, in turn, are applied to







In the FCOP1200 two-board firing package, the cross-hatched sections of the block diagram are located on the FCOA1200 board.

voltage comparators to produce phase reference inputs to the phase-locked-loop of the gate delay determinator circuit.

Gate Delay Determinator

The gate delay determinator is a six phase phase-locked-loop (PLL) consisting of a summing junction, low pass filter, voltage controlled oscillator (VCO) and two digital application-specific integrated circuit (ASIC) devices. The VCO frequency is locked to the power line frequency. The dc control voltage is injected into the summing junction to vary the thyristor gate delay angle.

Gate Drive Outputs

Twelve isolated and 30° spaced hard-firing thyristor gate pulse outputs are provided. The twelve gate command logic signals produced by the ASIC devices are applied to pulse amplifiers which actuate pulse transformers via pulse-shaping RC networks.

Electro-Mechanical Details

Board Configuration

The FCOG1200 is a single board configuration containing power supply, thyristor gate delay determinator logic, thyristor gate pulse transformers and associated circuitry.

The FCOP1200 is a two-board package. The FCOA1200 auxiliary firing board contains the power supply, thyristor gate pulse transformers and associated circuitry. (These items are shown as cross-hatched blocks in the functional block diagram.) The DDOG1200 gate delay determinator board is mounted on stand-offs above the FCOA1200. Interconnections between the two boards are made via ribbon cable.

Power Supply

The FCOG1200 power supply operates from an external 24Vac, 15VA, 1ø power source. The FCOP1200 supply requires 30Vac, 15VA, 1ø. Each power supply provides unregulated dc voltage and regulated 24, 12 and 5 Vdc.

Printed Circuit Boards

All printed circuit boards are fabricated from 2.36mm (.093") thick G10 epoxy-glass FR4 laminate by a UL registered fabricator.

Maximum Ratings

	FCOG1200	FCOP1200
AC Mains Voltage (Vac)	600	900
Pulse Transformer Hipot (Vac) -1 min	3500	5000
Gate Delay Command Voltage (Vdc)	-15 to +15	
Board Supply Voltage (Vac)	28	35
Available current (mAdc):		
+30V / +40 Vdc	300	
+24 Vdc	20	50
+12 Vdc	5	20
+5 Vdc	5	
Operating Temperature (°C)	0 to 70	
Storage Temperature (°C)	-55 to +150	

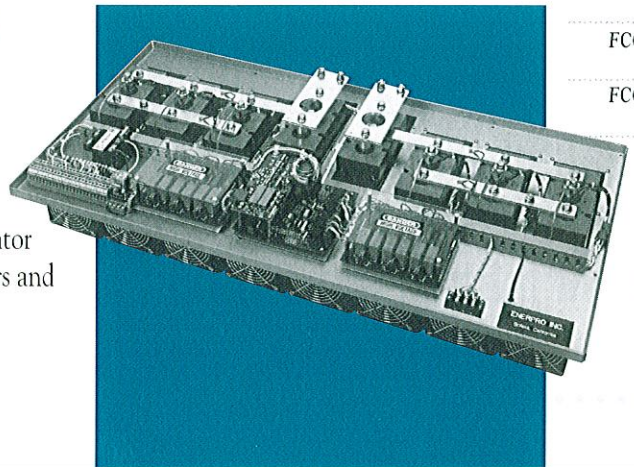
Electrical Characteristics

Gate Delay Command Voltage Range (Vdc)*	0 to 5	
Delay Angle Range (deg)*	6 to 174	
Phase Reference Angle Bias (deg)	0 or 30	
3dB Bandwidth (Hz)	167	
Delay Angle Tracking (deg):		
Same polarity	120 ± 0.5	
Different phase	180 ± 0.5	
Soft-Start Range (s)	0.05 to 5.0	
Power-Down Range (s)	0.05 to 10.0	
First Gate Pulse:		
Short Circuit Current (A)	1.8	1.2
First Gate Pulse, initial rate-of-rise (A/µs)	1.0	
Open Circuit Voltage (V)	14	32
Sustaining Gate Pulse:		
Short Circuit Current (A)	0.5	
Open Circuit Voltage (V)	7	15
Gate Pulse Width (µs):		
60 Hz	24 to 26	
50 Hz	20 to 22	
Gate Burst Width (deg)	120	
Number of Gate Pulses	384	

* or as customer specified

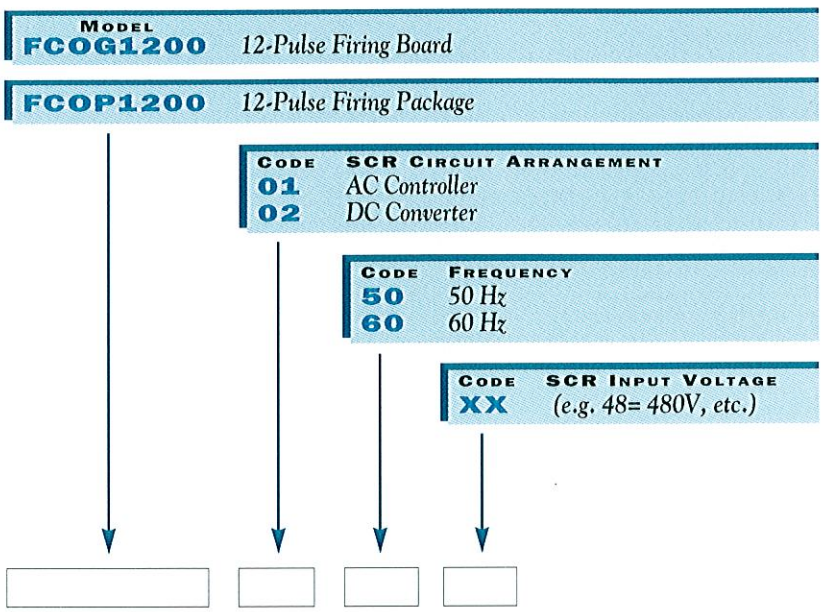
Mechanical Specifications

Printed Circuit Board Dimensions (mm / in)		
DDOG1200	n/a	152 x 191 6.0 x 7.5
FCOA1200	n/a	203 x 191 8.0 x 7.5
FCOG1200	191 x 194 7.5 x 7.625	n/a



ENERPRO can design and produce a complete 12-pulse system to your specifications, including SCRs, heat sinks, regulator boards and firing package like the 800A, parallel 12-pulse converter shown here.

Ordering Guide



Enerpro Applications Engineers are available by telephone or fax to answer any questions about single phase or three phase SCR control circuits or complete SCR assemblies.

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